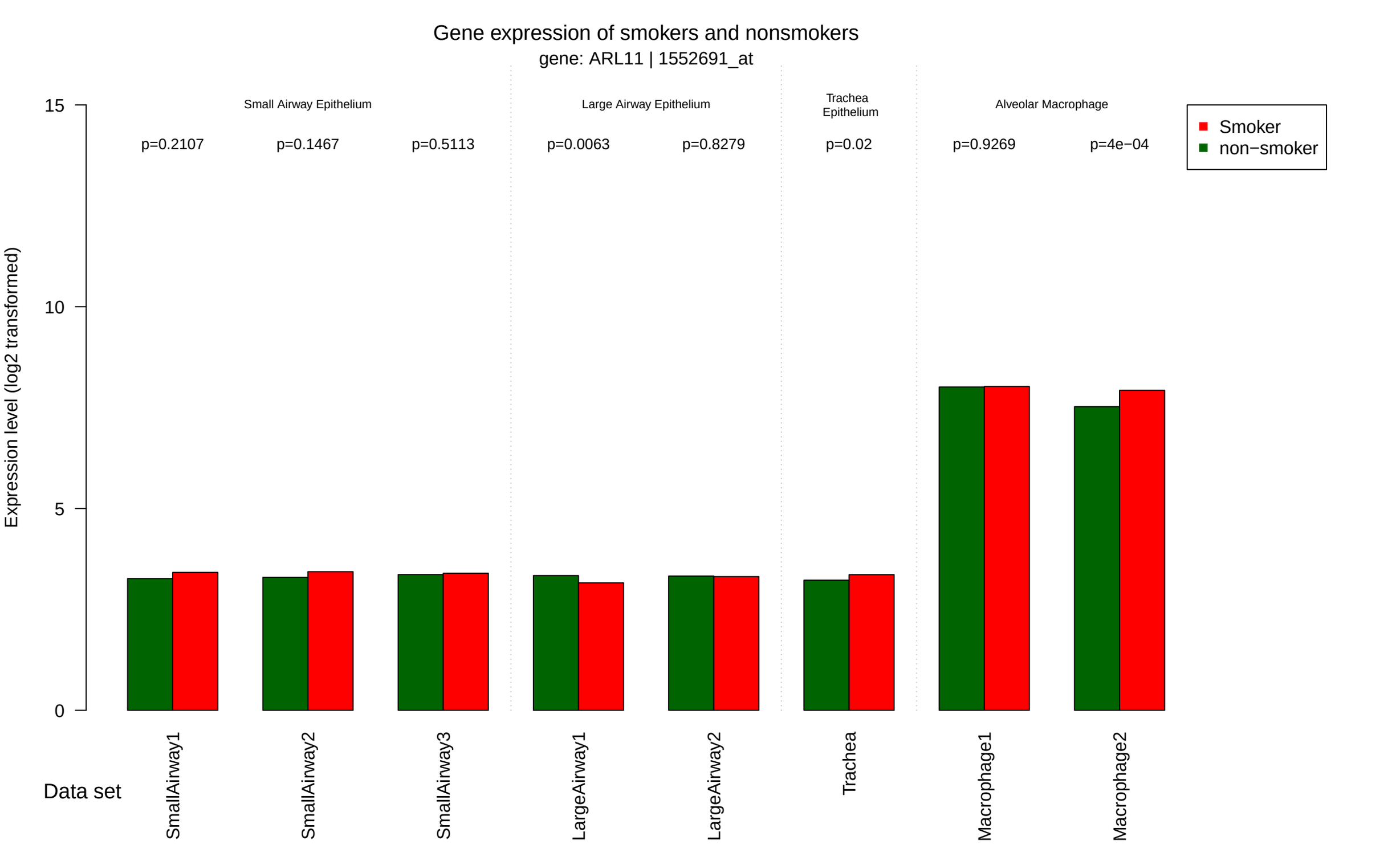
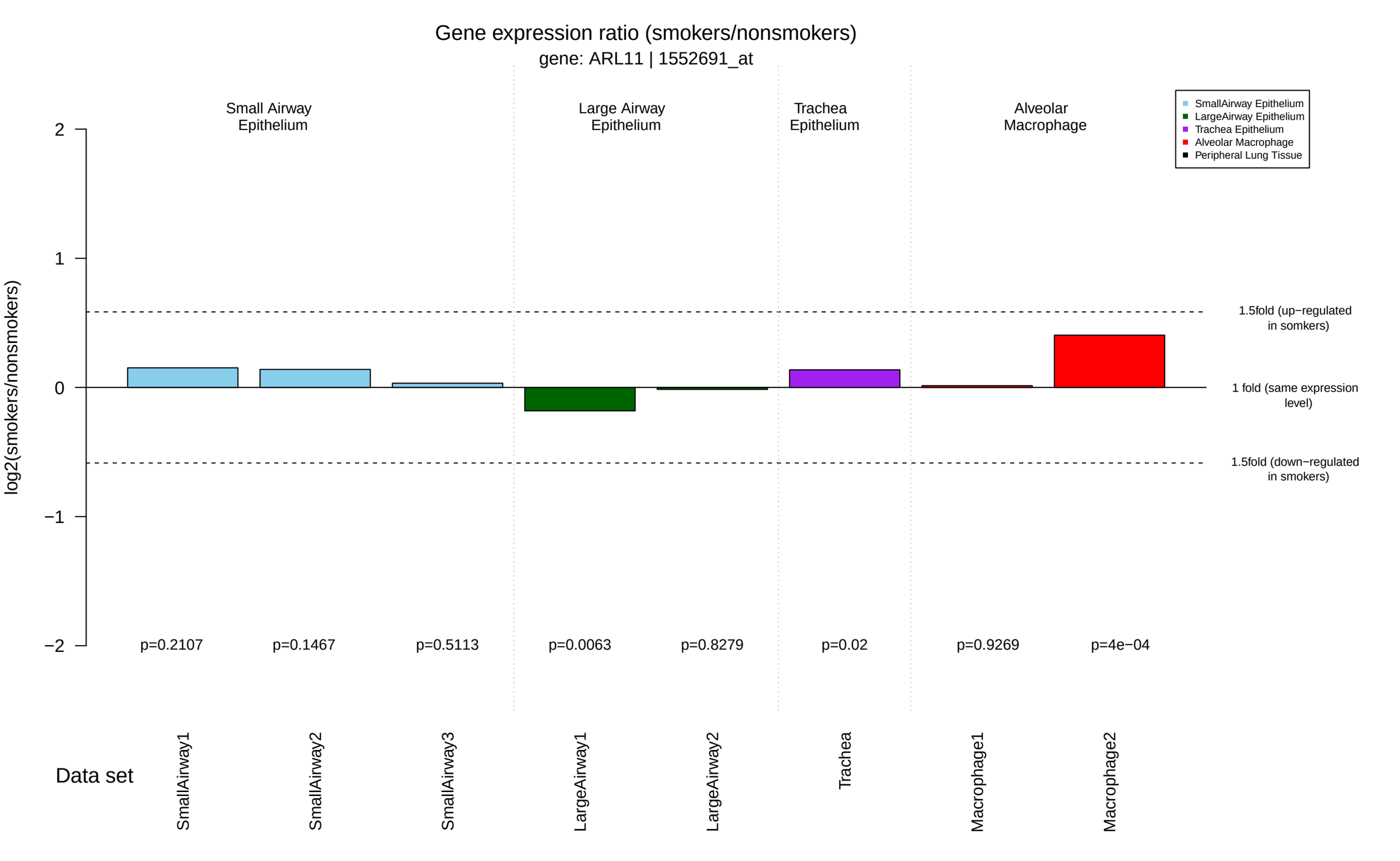
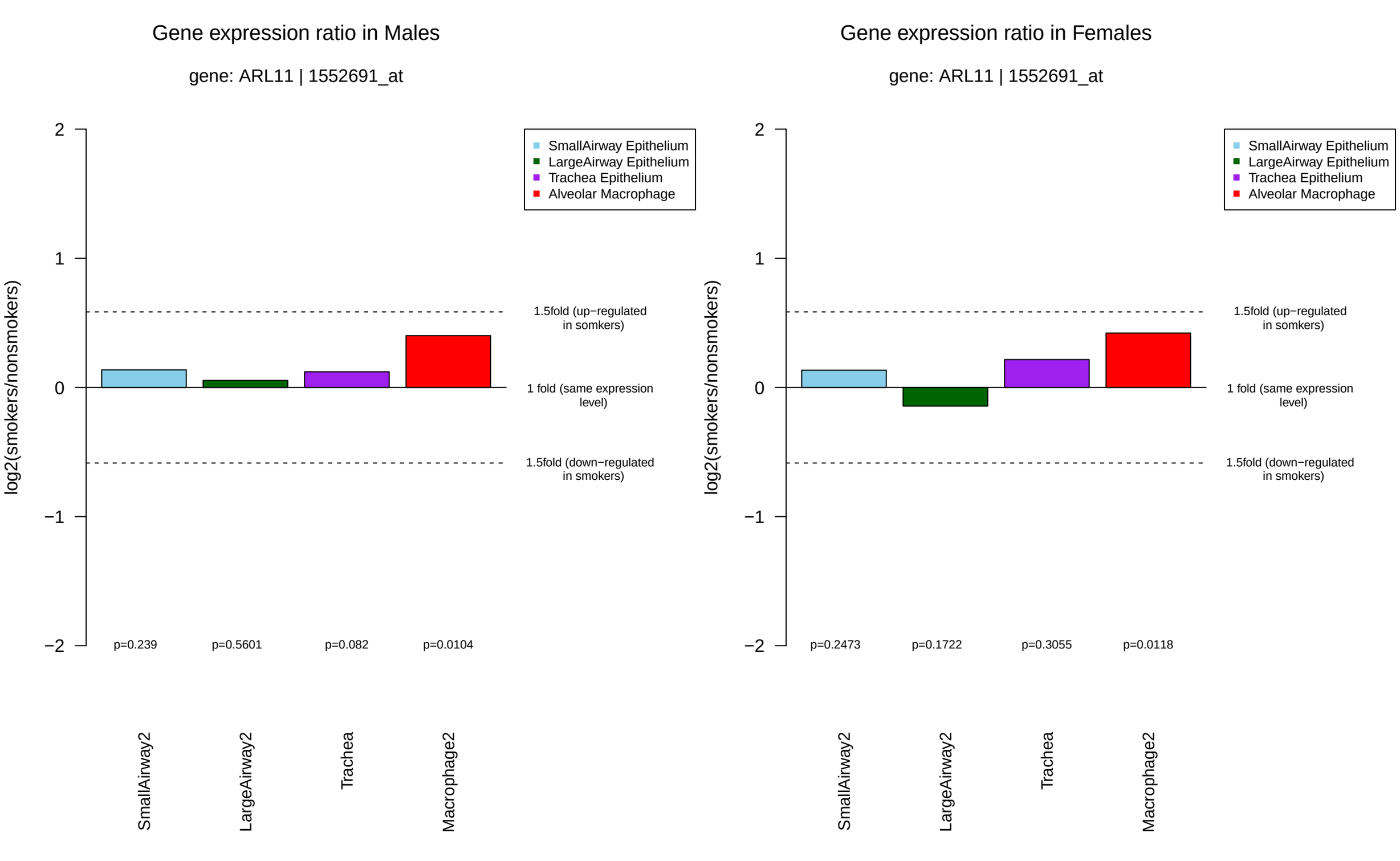
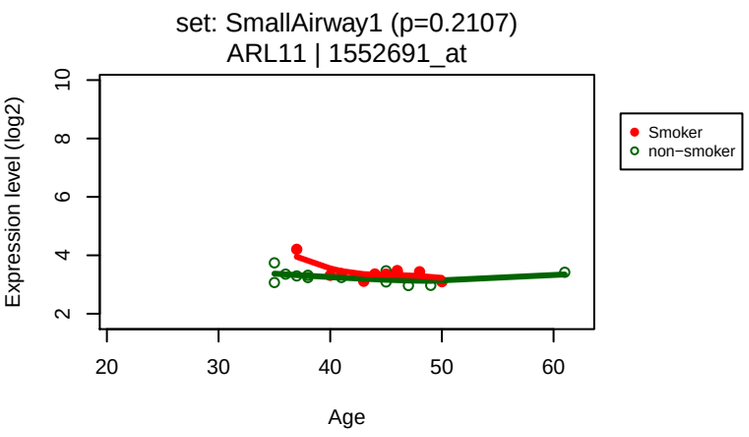
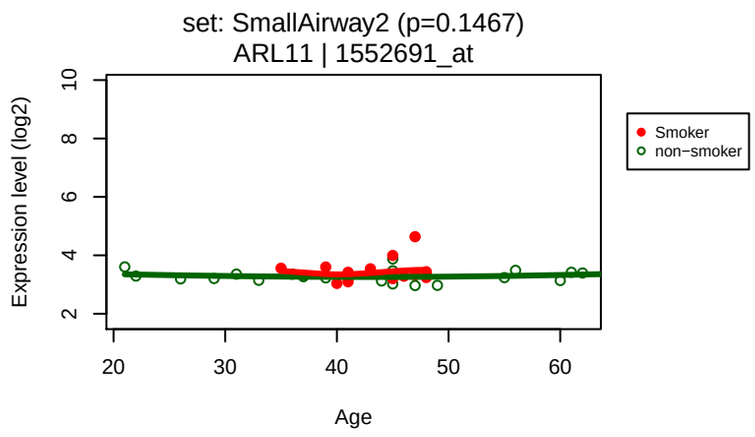
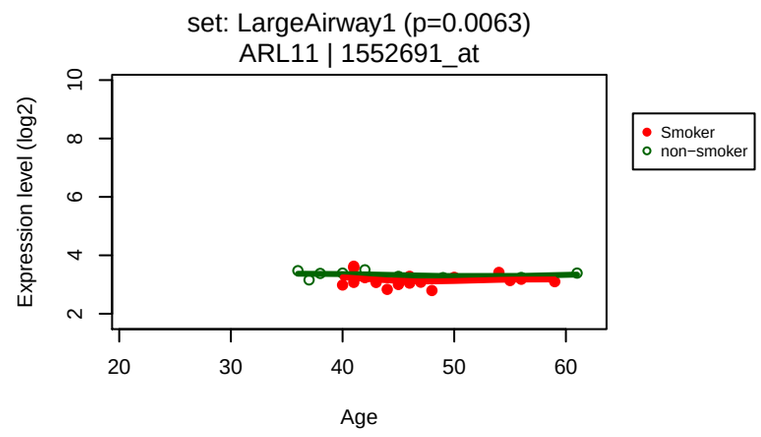

Gene expression of smokers and nonsmokers
gene: ARL11 | 1552691_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2107
p=0.1467
p=0.5113
p=0.0063
p=0.8279
p=0.02
p=0.9269
p=4e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

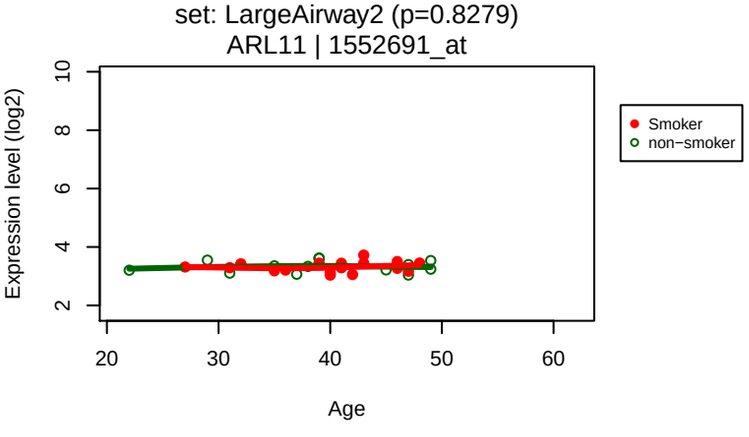

Gene expression ratio (smokers/nonsmokers)
gene: ARL11 | 1552691_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2107
p=0.1467
p=0.5113
p=0.0063
p=0.8279
p=0.02
p=0.9269
p=4e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

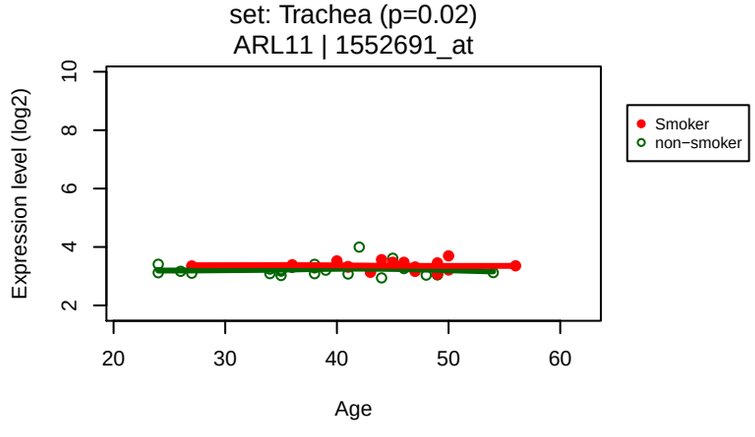

Gene expression ratio in Males
gene: ARL11 | 1552691_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.239
p=0.5601
p=0.082
p=0.0104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL11 | 1552691_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2473
p=0.1722
p=0.3055
p=0.0118
SmallAirway2
LargeAirway2
Trachea
Macrophage2

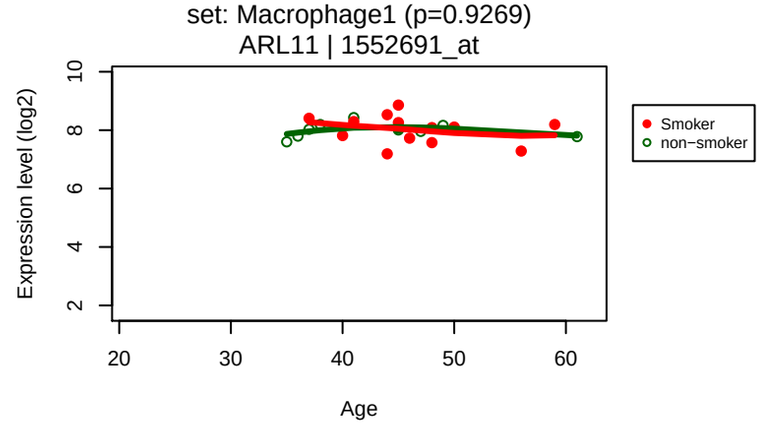

set: SmallAirway1 (p=0.2107)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker

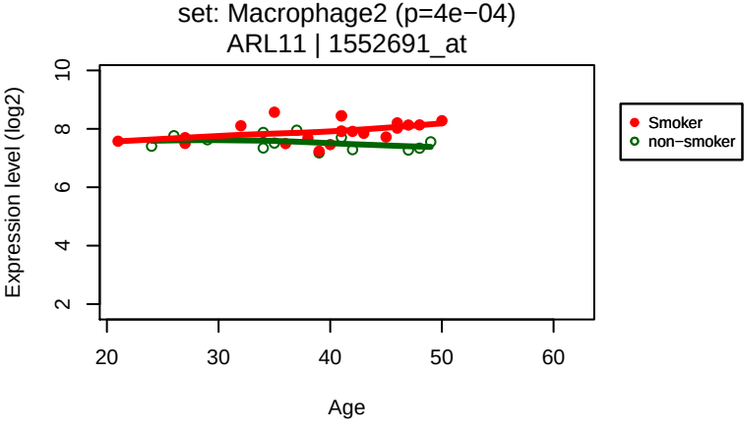

set: SmallAirway2 (p=0.1467)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker

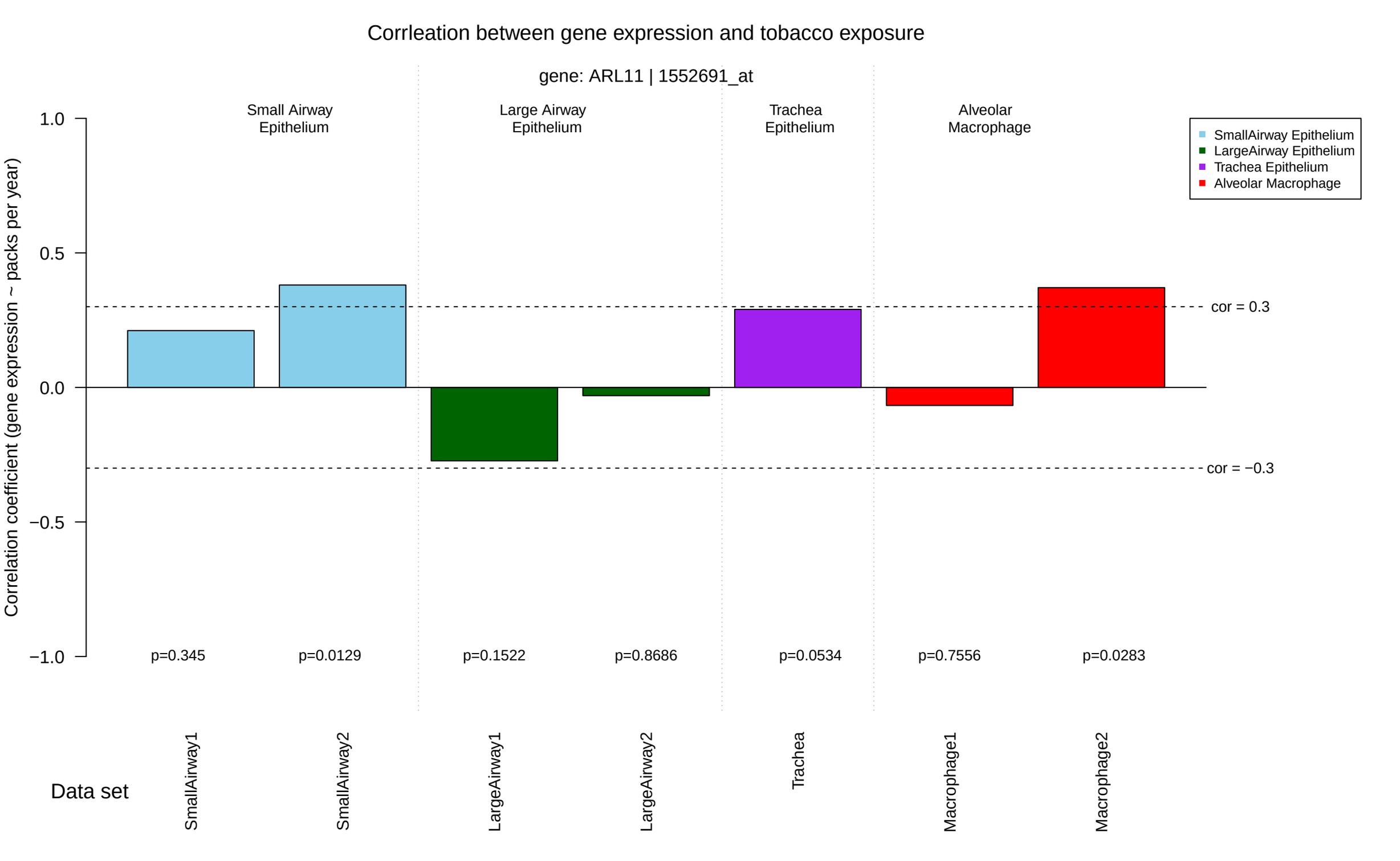

set: LargeAirway1 (p=0.0063)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8279)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.02)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9269)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=4e−04)
ARL11 | 1552691_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL11 | 1552691_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.345
p=0.0129
p=0.1522
p=0.8686
p=0.0534
p=0.7556
p=0.0283
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2